

# THE PLACE OF THE FAST REACTOR IN THE UNITED KINGDOM NUCLEAR PROGRAMME

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## 1. The Growth of Demand for Electricity

The demand for electricity in the United Kingdom has trebled since 1950, and has more than doubled in the last 10 years. This means that the electricity supply industry has had to contend with an average rise in demand of more than 8% per annum over the last 15 years. Throughout this period it has suffered from a shortage of capacity because, although estimates have been constantly revised upwards, actual demand has tended to exceed even the higher estimates. It is only in the last year or so that the rate of electricity growth has shown a tendency to level off.

These recent developments have caused some doubts about the growth of the industry in the future. It is clearly impossible for the extremely high rate of growth of the past to continue indefinitely while total energy demand grows at a much lower rate, and some observers maintain that the slightly slower rate of electricity growth recorded in the last few years is the first indication that its share of the total energy demand may soon reach its maximum and that before long its rate of growth may drop to the rate of growth of total energy demands.

Perhaps it is no coincidence that this slower rate of growth has occurred at a time when electricity prices have risen relative to gas prices. The electricity industry itself is confident that the 5 years covered by the National Plan will see a return to a higher rate of growth. Indeed, one expert<sup>(1)</sup> predicts an annual rate of growth in demand in the U.K. of over 8% until 1975. Table 1 sets out the figures of United Kingdom electricity sales in recent years, together with forecasts given in the National Plan and in the study of 1975 by the National Institute for Economic and Social Research.

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In an age of rapid technological change, such forecasts are clearly speculative, but better than no forecasts at all. Recent developments in the search for natural gas might suggest that some caution is called for in estimating future demand for electricity, but on the other hand, if relative prices are as significant as some commentators would apparently believe, the decline in electricity costs, relative to other costs, that the introduction of cheap nuclear power can bring could restore the 1950-60 rate of growth. Thus, it seems quite possible that by 1980 demand will be three times higher than in 1964, and about five times higher by 1990.

TABLE 1. ELECTRICITY SALES IN THE UNITED KINGDOM 1950 TO 1975<sup>(2)</sup>

| Year | Sales (thousand million<br>kilowatt hours) |
|------|--------------------------------------------|
| 1950 | 45.4                                       |
| 1955 | 67.4                                       |
| 1960 | 102.4                                      |
| 1964 | 140.2                                      |
| 1970 | 241.0                                      |
| 1975 | 332                                        |

These forecasts imply an increase in annual electricity demand per head of population from about 3000 kWh at present to over 7500 kWh in 1980, and nearly 12,000 kWh by 1990. These figures may seem startling at first sight, but it is interesting to compare them with the forecasts of two other major developed areas. U.S. electricity demand has like the U.K. doubled in the last 10 years, and annual demand per head has now reached 5500 kWh. Despite this high annual figure, the Federal Power Commission has recently published forecasts of U.S. electricity demand<sup>(3)</sup> which would involve a further doubling per head in the next 15 years.

In the European Economic Community, consumption per head is now just over two-thirds that in this country. It is expected by Euratom<sup>(4)</sup> to rise to 5600 kWh in 1980, and to 9500 kWh in 1990, so that the 2:3 ratio would remain roughly the same throughout the period, if both sets of forecasts were justified.

## 2. Nuclear Power as the Cheapest Source of Electricity

As the recent economic and technical appraisal of the advanced gas cooled reactor (AGR) to be built at Dungeness has demonstrated, nuclear power is now in a position to make an important contribution to meeting this future demand for electricity; for base load operation of stations commissioned in 1970 and later, nuclear power stations are confidently expected to be cheaper than stations using coal or oil as an energy source.<sup>(5)</sup>

The realization of the estimates quoted in the Fuel Policy White Paper will

ensure a valuable role for nuclear power in the 1970s, and further improvements are expected. From the early 'seventies therefore nuclear power will be competitive with conventional power at progressively lower load factors. Thus, if present trends for both nuclear power costs and for conventional fuel prices continue, most, if not all, of the net increase in electricity demand could after about 1975 be met most economically by the installation of nuclear generating capacity. It must be expected that there will continue to be a need for the installation of some new capacity especially to meet peak demands, and that the extremely low load factors of such plant will determine the choice in favour of minimum capital cost plant, even at the expense of high fuel costs and poor thermal efficiencies.

### **3. The Fast Reactor**

Not all of this new nuclear capacity, of course, will be provided by current systems; for example, the steam generating heavy water reactor, with its excellent fuel economy and good plutonium producing characteristics, might well in certain circumstances be an economic competitor to the AGR within a few years. Before 1980, however, if all goes well, the sodium cooled fast breeder reactor, for a prototype of which Government approval has now been given, should be commercially attractive, and should in due course become the predominant reactor system.

Other papers in this conference describe the U.K.'s fast breeder reactor development programme, outline the design features of the reactors under construction in the U.K. and explain the technical advantages which it is hoped the system will achieve, notably the ability of the reactor to breed more fissile material than it consumes, and thereby substantially improve the utilization of uranium as compared with any non-breeding system.

Because of these technical advantages, the fast reactor will have much lower fuel cycle costs than any thermal reactor can be expected to achieve. Its capital costs should, with further development work, be about the same, for a large reactor programme, as for thermal reactors available at the same time. The fast reactor promises, therefore, to be a cheaper source of electricity than any other known generating system.

It would, however, be unrealistic to assume that, as soon as a suitable design of fast reactor is available and has received sufficient confirmation from prototype operation, it would immediately replace all other systems for new stations. In the first place, though substantial quantities of plutonium are likely to be available from the earlier stations in the U.K. nuclear power programme, the initial inventory of plutonium for fast reactors is fairly high, and the rate of installation may be limited—for a time at least—by the supplies of plutonium available. In the long run, of course, when the fast reactor installation rate falls below the rate at which additional supplies of plutonium

are being bred by the fast reactor system (i.e. the doubling time of the fast reactor installation programme rises above the plutonium production doubling time), availability of plutonium will not be a problem, although by then growing demands for plutonium for overseas programmes of fast reactors may keep up the demand.

A further limitation on the rate of installation of fast reactors in the early years is likely to be the natural commercial caution of the various electricity authorities; these utilities will have the choice between a well-tried reactor system, the economic advantages of which compared with conventional fuels will by then be fully proven, and a new, potentially significantly cheaper, but largely untried reactor system. Their long-term interests will require them to test this new system because of its potential benefit to their costs of generation: it would be difficult, however, to blame them if they took the view that to devote one-quarter or one-third of their total new nuclear installation programme to the new system was the maximum reasonable until they had, by experience of construction and a prolonged period of prototype operation, satisfied themselves that the potential economic benefits were indeed realizable. This limitation is, however, essentially a short-term one.

#### **4. The Relationship between the Fast Reactor Design and the Development of the Programme**

Given then that the fast reactor is likely, because of its intrinsic fuel cycle cost advantage, to be significantly more economic than alternative systems, it is clearly desirable to obtain the maximum benefit possible from that advantage. Since the achievement of the full fuel cycle cost advantage of fast reactors depends on the availability of plutonium, it is clear that the size of that benefit and the best way of achieving it are to a considerable extent dependent on factors outside the fast reactor programme itself, i.e. the characteristics of the thermal (plutonium producing) reactors that have preceded the fast reactor in the power programme, the size and timing of build-up of the whole nuclear programme, and, indeed, the availability of plutonium from other sources. It will be seen therefore that there can be no unique answer to the question as to which is the most economic type of fast reactor, even given that such external factors as the rate of interest, price of uranium, etc., could be assumed to be known. In formulating, therefore, a development programme for the fast reactor it is necessary to consider, for example, how rapidly the rate of installation of fast reactors is likely to rise and the amount of plutonium likely to be available from other reactors. Only then can the importance, for example, of low initial inventories be assessed. Similarly, if we know what are the practical limits of the key fast reactor characteristics, we are in a better position to assess what thermal reactor characteristics are of greater importance, e.g. how important is a high rate of plutonium production. This assess-

ment will be a continuing process as the size and timing of the whole nuclear power programme gradually becomes more definite; it may also help to indicate how over a period the optimum design of a fast reactor will change.

### 5. The Installation Programmes

To assess these effects, to consider what the contribution of fast reactors to the U.K. power programme might be, and to estimate how variations in

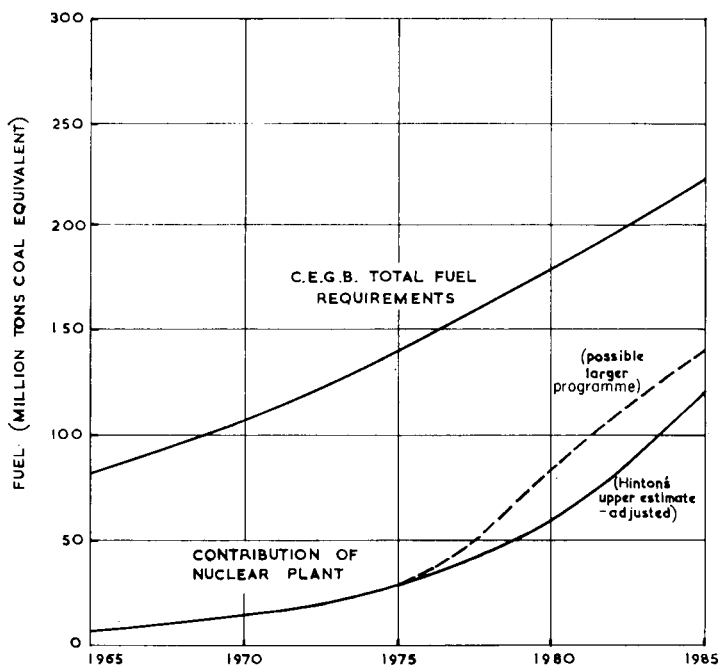


FIG. 1. Estimated Central Electricity Generating Board total fuel requirements to 1975 with tentative extensions to 1985 and with the exploration of the nuclear contribution

reactor parameters could affect the position, it is necessary to make some assumptions on the likely rate of installation of nuclear power plant. Taking account of the views of the rate of growth of electricity demand expressed earlier, one useful indication of the possible introduction of nuclear power is given in Fig. 1, based on the graph with which Lord Hinton illustrated his Citrine Lecture delivered to the British Electrical Power Conference last year.

Lord Hinton's lecture, however, was prepared over 12 months ago, well before the assessment of tenders for the Dungeness "B" station was complete; that assessment, and other factors such as the Government's decision to increase the planning figure for the Second Nuclear Power Programme from

5000 MW to 8000 MW, has led to increased confidence in the place that nuclear power will have in meeting the electricity demand of the future. It might, therefore, now be reasonable to regard Lord Hinton's large programme as a moderate estimate of nuclear capacity and to allow for the possibility that the rate of nuclear installation might be larger than his estimate. A possible

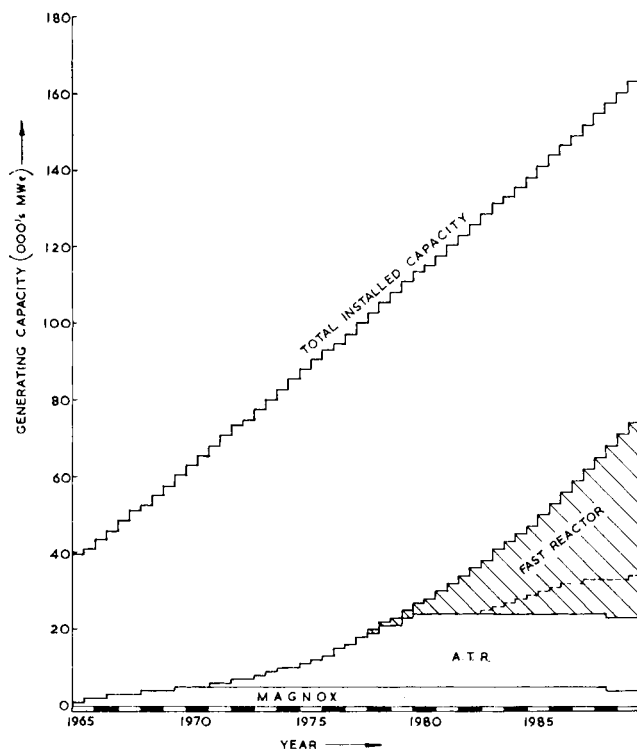


FIG. 2. Smaller notional power programme.

larger programme, which, however, can be shown on certain assumptions of future demand, etc., to be well within the maximum possible, is also indicated in Fig. 1.

The curves of Fig. 1 can be translated into approximate capacity figures, which can in turn be further extrapolated beyond 1985. These figures should be treated as illustrative only; they obviously depend on a number of assumptions, e.g. about the rate of growth of demand for electricity, the nature of the country's electricity generating system in 20–25 years' time and the competition offered to nuclear power by conventional energy sources, some of which could turn out to be wide of the mark. The results of this extrapolation, and the breakdown of the nuclear sector are set out in Figs. 2 and 3. The shaded

parts of these figures show the maximum fast reactor programmes (given no limitation on the availability of plutonium) after some account has been taken of the limiting factor mentioned in section 3.

The magnox component of the nuclear sector is well established. The successive thermal reactor has been denoted generically as an advanced

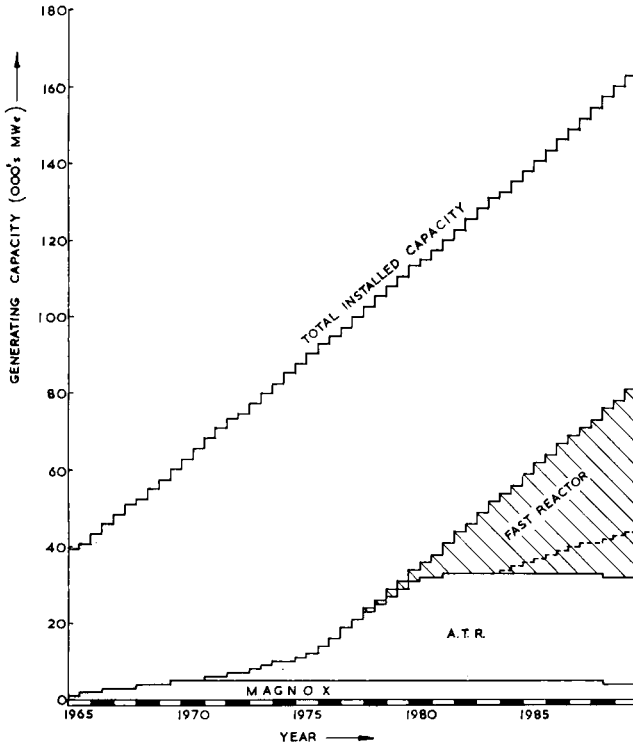


FIG. 3. Larger notional power programme.

thermal reactor (ATR), although initially it will be the AGR. In general, availability of plutonium may limit the fast reactor installation to rather less than that shown shaded in Figs. 2 and 3. If it did, the installation of ATR's would be increased at the expense of that of fast reactors, typically as shown by the dotted line in the figures.

## 6. Assessment of the Fast Reactor's Place in the System

Within a closed system, to which external supplies of plutonium are not available, the installation of fast reactors will be limited by the availability of plutonium from reactors already in operation in the generating system. One of the authors has recently carried out a study indicating for two, rather

cautious, illustrative nuclear power programmes how the fraction of electricity generated by fast reactors can be deduced from the technical parameters of the relevant reactor systems.<sup>(6)</sup> As different assumptions on, for example, fast reactor fuel inventory or thermal reactor plutonium production are taken, so the fraction of electricity generated by fast reactors will change.

For present purposes, consideration may be limited to the "system" defined by the greatest part of the power programme that could be affected by

TABLE 2

|                                                                                   |                                                              |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------|
| Maximum load factor                                                               | 80%                                                          |
| Proportion of total installed capacity @ max LF:                                  | 1976 30%                                                     |
|                                                                                   | 1990 40%                                                     |
| Load factor of capacity exceeding this proportion:                                | falling uniformly to 10% @ 70.5% of total installed capacity |
| Discounted electricity generation from "system" (as defined in Appendix I), $U$ : |                                                              |
| Smaller programme                                                                 | 246,770 MW(e)-yr*                                            |
| Larger programme                                                                  | 237,930 MW(e)-yr*                                            |
| Plutonium production from magnox station                                          | 0.548 kg $\equiv$ Pu 239/MW(e)-yr                            |
| Plutonium production from ATR                                                     | 0.1, 0.2 or 0.3 kg $\equiv$ Pu 239/MW(e)-yr                  |
| Fast reactor inventory @ 80% LF                                                   | 2, 3 or 4 kg $\equiv$ Pu 239/MW(e)                           |
| Fast reactor linear doubling time @ 80% LF                                        | 5, 10, 15 or 20 years                                        |
| Proportion of fast reactor inventory outside core @ 80% LF                        | 0.333                                                        |
| Nuclear station lifetime                                                          | 25 years                                                     |
| Discount rate                                                                     | 7½ %/year                                                    |
| Load factor order of merit                                                        | magnox, fast reactor, ATR                                    |
| Base date                                                                         | start 1976                                                   |
| Plutonium allocated to civil nuclear power from                                   | mid-1970                                                     |

\* The two nuclear programmes have been chosen not only to illustrate differences of size, but also different rates of nuclear build-up. The smaller programme builds up more gradually, whereas the larger programme grows very quickly in the period 1976-81. Consequently the total fast reactor installation is similar in both cases, and the values of  $U$  are close together.

the fast reactor, as indicated by the shaded areas in Figs. 2 and 3. The electricity generated\* by fast reactors may then be expressed as a fraction  $x$  of the total electricity generated by this "system". This fraction has been obtained for the various alternative parameters set out in Table 2.

In order to allow for the differing isotopic composition of plutonium discharged by different reactors, all plutonium is expressed as its "equivalent Pu 239", having the same reactivity worth in fast reactors. The fast reactor inventory is made up of the plutonium in the reactor core, and in the fresh and spent fuel hold-up both at the station and at the fuel fabrication and

\* In this context electricity generated is obtained as the sum, over the period of time considered, of the annual quantities each discounted back to the base date at the appropriate discount rate.

reprocessing plants. Earlier work has shown that results depend only slightly on the way in which the inventory is proportioned between plutonium in and

TABLE 3. SMALLER PROGRAMME

| FR inventory<br>@ 80% LF<br>(kg $\equiv$ Pu 239/<br>MW(e)) | FR linear<br>doubling time<br>@ 80% LF<br>(yr) | Fraction of electricity generated by fast reactors (x) |       |       |
|------------------------------------------------------------|------------------------------------------------|--------------------------------------------------------|-------|-------|
|                                                            |                                                | ATR Pu prodn.<br>(kg $\equiv$ Pu 239/MW(e)-yr) = 0.1   | 0.2   | 0.3   |
| 2                                                          | 5                                              | 0.880                                                  | 0.975 | 1.000 |
|                                                            | 10                                             | 0.759                                                  | 0.890 | 0.983 |
|                                                            | 15                                             | 0.689                                                  | 0.842 | 0.946 |
|                                                            | 20                                             | 0.654                                                  | 0.807 | 0.920 |
| 3                                                          | 5                                              | 0.685                                                  | 0.788 | 0.871 |
|                                                            | 10                                             | 0.547                                                  | 0.677 | 0.785 |
|                                                            | 15                                             | 0.490                                                  | 0.620 | 0.735 |
|                                                            | 20                                             | 0.468                                                  | 0.594 | 0.703 |
| 4                                                          | 5                                              | 0.562                                                  | 0.659 | 0.740 |
|                                                            | 10                                             | 0.424                                                  | 0.539 | 0.640 |
|                                                            | 15                                             | 0.381                                                  | 0.488 | 0.586 |
|                                                            | 20                                             | 0.359                                                  | 0.466 | 0.560 |

TABLE 4. LARGER PROGRAMME

| FR inventory<br>@ 80% LF<br>(kg $\equiv$ Pu 239/<br>MW(e)) | FR linear<br>doubling time<br>@ 80% LF<br>(yr) | Fraction of electricity generated by fast reactors (x) |       |       |
|------------------------------------------------------------|------------------------------------------------|--------------------------------------------------------|-------|-------|
|                                                            |                                                | ATR Pu prodn.<br>(kg $\equiv$ Pu 239/MW(e)-yr) = 0.1   | 0.2   | 0.3   |
| 2                                                          | 5                                              | 0.885                                                  | 0.985 | 1.000 |
|                                                            | 10                                             | 0.825                                                  | 0.957 | 1.000 |
|                                                            | 15                                             | 0.771                                                  | 0.947 | 1.000 |
|                                                            | 20                                             | 0.733                                                  | 0.921 | 1.000 |
| 3                                                          | 5                                              | 0.715                                                  | 0.825 | 0.894 |
|                                                            | 10                                             | 0.604                                                  | 0.760 | 0.865 |
|                                                            | 15                                             | 0.546                                                  | 0.710 | 0.841 |
|                                                            | 20                                             | 0.519                                                  | 0.680 | 0.819 |
| 4                                                          | 5                                              | 0.595                                                  | 0.706 | 0.782 |
|                                                            | 10                                             | 0.468                                                  | 0.616 | 0.723 |
|                                                            | 15                                             | 0.422                                                  | 0.559 | 0.680 |
|                                                            | 20                                             | 0.400                                                  | 0.532 | 0.652 |

outside the reactor core. Delays arising from approach to fuel cycle equilibrium, from fabrication and reprocessing of generated plutonium and from loading cores of new reactors are accounted for separately.

The fraction  $x$  is shown in Tables 3 and 4 for various combinations of reactor parameters and for the two nuclear-power programmes indicated in Figs. 2 and 3, but extrapolated to the year 2000 in order to provide a reasonable assessment of fast reactors. (Some of the values shown in these tables are in fact unlikely to be achievable with a sodium cooled FR.) The tables show that there are no critical values of either inventory or doubling time. The fraction  $x$  continually increases, with accompanying economic gains, as both inventory and doubling time are reduced. In the circumstances assumed, the same increase in electricity generation from fast reactors could be achieved either by a reduction in doubling time from 20 to 5 years or by a reduction of 15–35% on the inventory.

### 7. Sodium Cooled Fast Reactors with Mixed Oxide Fuel

In the U.K., fast reactor development is mainly concentrated on mixed oxide fuel with sodium cooling and it is therefore of interest to examine the lowest doubling time which this system can achieve. Inclusive of reprocessing losses, the highest breeding gain attainable in this system is about 0.40 atoms  $\equiv$  Pu 239/fission. With a station efficiency of 40% and a heat output of 1 MW(h)D/g fissioned, the doubling time at 80% load factor becomes 14 years for an inventory of 4 kg  $\equiv$  Pu 239/MW(e). On this basis, lower doubling times can only be achieved by reducing the inventory. Breeding limitations thus force a dependence of doubling time on inventory, although the two quantities are independent from the economic aspect.

A reduction in inventory demands a higher mass rating of the fuel, which may be achieved by reducing the diameter of the fuel pin and increasing the temperature rise within it. The temperature rise itself is determined by the linear rating which experience to date indicates may well be as high as 500 W/cm for mixed oxide fuel. If unduly fine clearances are to be avoided, the minimum practicable diameter of fuel pin is about 0.2 in., giving a mass rating of about 300 W/gm heavy atoms. The concentration of equivalent Pu 239 is typically 15% so that a station efficiency of 40% results in a core loading of 1.50 kg  $\equiv$  Pu 239/MW(e). With the same hold-up fraction as in Table 2 the inventory at 80% load factor becomes 2.25 kg  $\equiv$  Pu 239/MW(e). If this inventory could be coupled with the maximum possible breeding gain, then the doubling time of 14 years quoted above for an inventory of 4 kg  $\equiv$  Pu 239/MW(e) would become only 8 years.

These considerations suggest that mixed oxide fuel cooled with sodium should be able to achieve at least 80% of the maximum possible advantage from such fast reactors and that with a larger pre-1978 programme of thermal reactors, or with a high proportion of good plutonium producing thermal reactors, 100% is possible. The precise saving in generating cost to the "system" that the fast reactor can bring can only be calculated by a detailed

system analysis in which the optimum year-by-year costs of an integrated system of thermal and fast reactors are compared with the costs of a corresponding all-thermal reactor system in which all the plutonium produced is recycled and thermally burned. However, for the purpose of this paper, an indication of the order of magnitude of the possible saving can be obtained, as is shown in Appendix I, from the fraction  $x$  and the generating cost differential between the thermal and fast reactors, due allowance being made for the recycling of plutonium in thermal reactors as an alternative to its use in fast reactors. The possible magnitude of this differential is discussed in Appendix II, from which it would appear that the saving in "system" generating cost can hardly be less than 0.40 mills/kWh; a more probable figure would be one some 50% higher, say, 0.70 mills/kWh. Even higher figures have also been quoted, and could well be realized if uranium prices rose sharply.

### 8. Conclusion

On the basis of this analysis which, while illustrative, is considered to be reasonably realistic, and indeed cautious, in its assessment of the potential role of the fast reactor, it is possible to indicate the potential economic gains from the incorporation of the fast reactor in the generating system of this country. On the basis of the minimum likely differential between thermal and fast reactor generating costs described in Appendix II, which as that Appendix suggests is likely to be linked with an  $x$  value of 1.0 or nearly 1.0, the savings in expenditure for the illustrative programmes chosen expressed as their present worth in 1976 would exceed £300m. On more likely assumptions but still at relatively low uranium prices, and allowing for a somewhat smaller  $x$  value, these savings would become £500m. Were uranium prices to double—by no means an unrealistic assumption if the fast reactor is not incorporated in the electricity generating programme—the benefit from fast reactors would increase by about 50%—to say £750m. Whether the very low electricity generation costs foreseen as a result of these developments in nuclear power will themselves significantly increase the demand for electricity in the 1980s and 1990s it is impossible now to say: if they did, then the economic gains would again be larger than those related to the illustrative programmes we have adopted here. Clearly, if fast reactors can indeed achieve the performance that is now being promised from them, their place in the U.K. nuclear power programme seems assured.

### References

1. G. F. RAY in BECKERMAN *et al.*, *The British Economy in 1975*, National Institute for Economic and Social Research, 1965, p. 312.
2. *The Ministry of Power Statistical Digest 1964*, table 73. *The National Plan* (Cmnd. 2764), pp. 11–42.  
BECKERMAN *et al.*, *op. cit.*, p. 317, table 10.17.

3. Federal Power Commission—National Power Survey 1964.
4. H. MICHAELIS, *Situation and Trends of Nuclear Energy in the European Atomic Energy Community*, Euratom, 1964.
5. White Paper on *Fuel Policy*, Cmnd. 2798, p. 23.
6. C. E. ILIFFE, *Economics of Plutonium Breeding, Plutonium Inventory and Uranium Conversion*, UKAEA TRG Report 1207 (R). (Obtainable from HMSO.)

## Appendix I

### Economic Assessment of Fast Reactors

The method of economic assessment used here is similar to the one described in Reference 6. In that work, however, the load factors of all nuclear stations were assumed to be the same at all times whereas in the present instance they may vary. The consequent modification to the method, however, is simple and straightforward.

As in Reference 6, consideration may be limited to a "system", defined by the greatest part of the power programme that would be affected by fast reactors. If only changes in fast reactor specification are to be assessed, the system becomes merely the fast reactor sector of the programme when plutonium supplies are unlimited. Discounted to any convenient "base" date, the electricity generated by this system may be written  $U$ .

In general, plutonium availability will restrict the installation of fast reactors so that their discounted electricity generation becomes less than  $U$ , say  $xU$ . On division by the corresponding quantity when all the fast reactors operate at 100% load factor, there results the fast reactor mean load factor,  $L_f$ .

The average system generating cost per unit sent out may be written  $C$ , and the station generation cost  $c$ , with suffix  $a$  for ATR and  $f$  for fast reactor. A further suffix will denote the appropriate (discounted mean) load factor. For the purpose of the model, it is assumed that all reactors of the same type have the same costs of construction, irrespective of the date of installation, and the same costs of fuelling per unit of electricity generated. In this closed "system" of nuclear stations any price assigned to plutonium cancels out in transactions between thermal and fast reactors. The costs  $c$  are therefore quoted free of any credit or debit for plutonium. Provided that the ATR and fast reactor are next to one another in the load factor order of merit (an oversimplification which can, however, be shown to be justified in the circumstances considered) then the reduction in "system" generating cost through the introduction of fast reactors is simply:

$$\Delta C_f = (c_a - c_f)_{L_f} x \quad (1)$$

and the corresponding reduction in discounted expenditure is:

$$\Delta X_f = U \Delta C_f. \quad (2)$$

Whatever change is made to the fast reactor specification, it is implicit in formulae (1) and (2) that there is no change in either the specification or the fuel cycle of the ATR. Whilst this implication is essential in assessing changes in the fast reactor itself, it is not necessarily appropriate to an assessment of the advantage of introducing fast reactors at all. Such an assessment can only be made by considering all the reactor systems alternative to fast reactors. In the present instance, however, the fast reactor will be compared with the alternative system in which the plutonium from thermal reactors, instead of being fed to fast reactors, is fed to ATR's.

If there are ATR's available for operation on a plutonium feed in excess of those capable of replacement by fast reactors,\* then the system must be extended to include these ATR's. As a fraction of the extended system the extension may be written  $\lambda$ , so that the discounted electricity generation from the extended system becomes:

$$U/(1 - \lambda).$$

The corresponding mean load factor may be written  $L$ .

Deferring consideration of plutonium from stations outside the extended system,† all ATR's on a plutonium feed will be assumed in the first instance to operate on plutonium recycle with a makeup of enriched uranium. Denoting such stations by the suffix ru, their introduction would reduce the average generating cost of the extended system by an amount:

$$\Delta C_{ru} = (c_a - c_{ru})L. \quad (3)$$

The corresponding reduction in discounted expenditure is then:

$$\Delta X_{ru} = \frac{U}{1 - \lambda} \Delta C_{ru}. \quad (4)$$

However, a number of these ATR's could be supplied with a makeup of plutonium from the stations outside the extended system instead of enriched uranium. Such stations will be denoted by the suffix rp, and their mean load factor written  $L_{rp}$ . If the discounted electricity generation from these stations is a proportion  $x_r$  of that from the extended system, then analogous to formula (1), the consequent reduction in the generating cost of the extended system would be:

$$\Delta C_{rp} = (c_{ru} - c_{rp})L_{rp} x_r \quad (5)$$

and the corresponding reduction in discounted expenditure:

$$\Delta X_{rp} = \frac{U}{1 - \lambda} \Delta C_{rp}. \quad (6)$$

\* ATR's outside the "system" as defined in the second paragraph of this Appendix.

† e.g. from the magnox stations, and from any ATR's not capable of operating on a plutonium feed.

Combining the above results, the net saving in discounted expenditure through the introduction of fast reactors becomes:

$$\begin{aligned}\Delta X &= \Delta X_f - (\Delta X_{ru} + \Delta X_{rp}) \\ &= \left[ (c_a - c_f)_{L_f} x - \frac{(c_a - c_{ru})_L + (c_{ru} - c_{rp})_{L_{rp}} x_r}{1 - \lambda} \right] U.\end{aligned}\quad (7)$$

Referred to the original system, that is the maximum possible fast reactor installation with unrestricted supplies of plutonium, the corresponding net reduction in average generating cost is:

$$\begin{aligned}\Delta C &= \Delta X / U \\ &= (c_a - c_f)_{L_f} x - \frac{(c_a - c_{ru})_L + (c_{ru} - c_{rp})_{L_{rp}} x_r}{1 - \lambda}.\end{aligned}\quad (8)$$

In practice, as already noted, it will be necessary to allow for variations in both the costs and the technical characteristics of the various types of ATR that will be interposed between the magnox stations and the fast reactor, but the same principles can be applied.

## Appendix II

### The Differential in Generating Cost between Thermal and Fast Reactors

The differential in generating cost between thermal and fast reactors in a closed system attributing no value to plutonium will reflect differences in

- (i) capital cost,
- (ii) cost of fuel fabrication and reprocessing,
- (iii) cost of uranium.

To assess the *minimum* advantage the fast reactor might have, we need to determine how small such a differential could conceivably become. For the purpose of this analysis it is perhaps legitimate to simplify the pattern by making two important assumptions which eliminate from consideration (i) and (ii) above. First, as reactor development proceeds, and as the size of power stations increases, differences in capital cost of different reactor systems are likely to be small. Second, fast reactor burnup is likely to be much greater (perhaps several times) than the burnup obtainable in thermal reactors, and it seems reasonable to assume that the probable greater cost in fabrication and reprocessing of the higher burnup fuel will be fully balanced by this greater burnup. On these assumptions the cost of uranium (incurred only by thermal reactors) will be the main factor in determining the differential between thermal and fast reactors.

The lowest uranium cost will be achieved by thermal reactors giving the greatest neutron economy. Such reactors may never in practice be installed

to any appreciable extent since they generally suffer overriding disadvantages in either capital cost or burnup or both. The best alternative to a fast reactor may thus in fact be one with relatively poor neutron economy, though even then the differential in cost owing to (i) and (ii) above cannot be large. However, if one seeks a reactor with a good neutron economy which avoids the extremes of high capital cost and low burnup, a possible choice will be the non-superheat version of the steam generating heavy water reactor (SGHWR). Typically, the uranium costs (at current U.S. prices for enriched uranium) of such a reactor might be:

|                                          |
|------------------------------------------|
| 0.64 mills/kWh replacement charge        |
| 0.09 mills/kWh inventory charge @ 80% LF |
| 0.01 mills/kWh hold-up charge            |
| <hr/> 0.74 mills/kWh uranium cost        |

Detailed studies have been carried out to determine the extent to which this generating cost differential might be reduced by the recycling of plutonium in ATR's characterized as above and by the feeding to such reactors of the plutonium produced in magnox stations and in ATR's for which operation on a plutonium feed might not be appropriate (e.g. ATR's commissioned before 1975). These have, necessarily, had to take into account such factors as the effect of the approach to recycle equilibrium and the complicated pattern of supply and demand for plutonium feed material. They have indicated that the optimum re-use of plutonium in such ATR's might reduce the saving in "system" generating cost by about 0.35 mills/kWh; such a figure typically represents the net effect of the second term in formula (8) of Appendix I. Since the load factors  $L_f$ ,  $L$  and  $L_{rp}$  are all found to lie between 0.782 and 0.800, it is appropriate to set  $(c_a - c_f)_{L_f} = 0.74$  mills/kWh. On this basis, a minimum saving in "system" generating cost due to introducing fast reactors would appear to be about 0.40 mills/kWh.

While there is no necessary connexion between the good neutron economy, necessary on the assumption above to produce the minimum differential between fast and thermal reactor generation costs in a closed "system", and a high plutonium production rate—which, as Tables 3 and 4 show, is necessary to permit large values of  $x$ —it is not unreasonable to assume that a reactor with good neutron economy could, if desired, be adapted to become a good plutonium producer. Looked at another way it is difficult to see how, if the plutonium production rate of the ATR adopted is low, the neutron economy can be good enough to produce the sort of figures outlined above.

The above analysis has deliberately sought to avoid detailed cost comparisons of alternative systems: the aim has been to describe a method, to indicate the interaction between the various characteristics of alternative reactor systems and the importance of a system wide analysis, and on the basis of what are believed to be realistic and cautious assumptions to draw some

broad conclusions. A number of such detailed cost comparisons have, however, been done, e.g. Paper 3/6: these will give a range of actual calculated cost differentials, based on particular assumptions, which appear to indicate a cost advantage to the fast reactor greater than the minimum quoted above. To the extent therefore that this Appendix has underestimated the differential likely in practice to emerge, the benefit from fast reactor development has been understated.